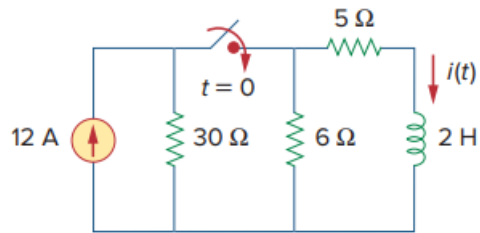


Problem

The switch in Fig. 7.71 was open for a long time but closed at $t = 0$. If $i(0) = 10\text{A}$, find $i(t)$ for $t > 0$ by hand and also by *PSpice*.

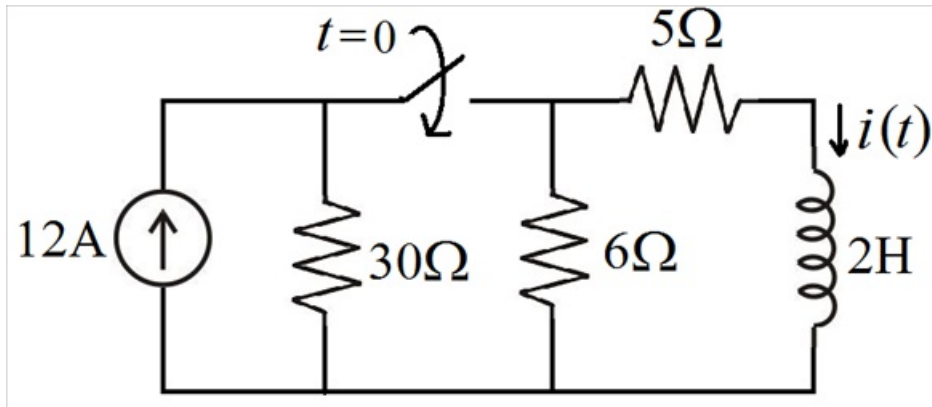
Fig. 7.71:



Step-by-step solution

Step 1 of 13

Given circuit diagram

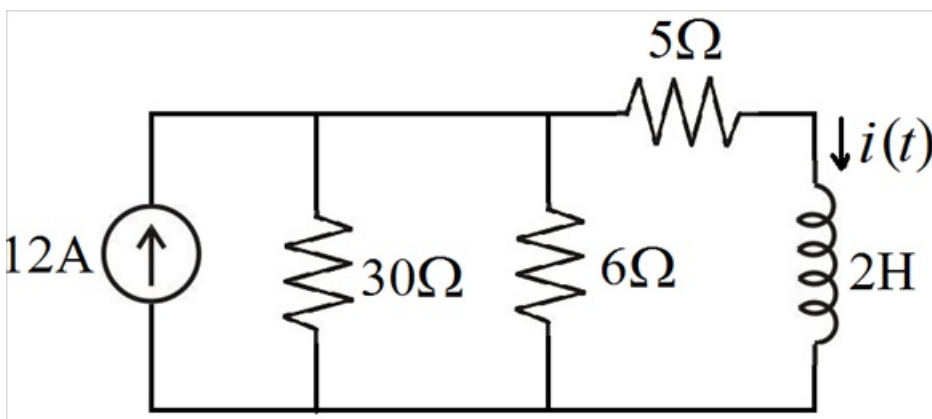


Step 2 of 13

Given $i(0) = 10\text{A}$

For $t > 0$ the switch is closed and we get the RL network as below

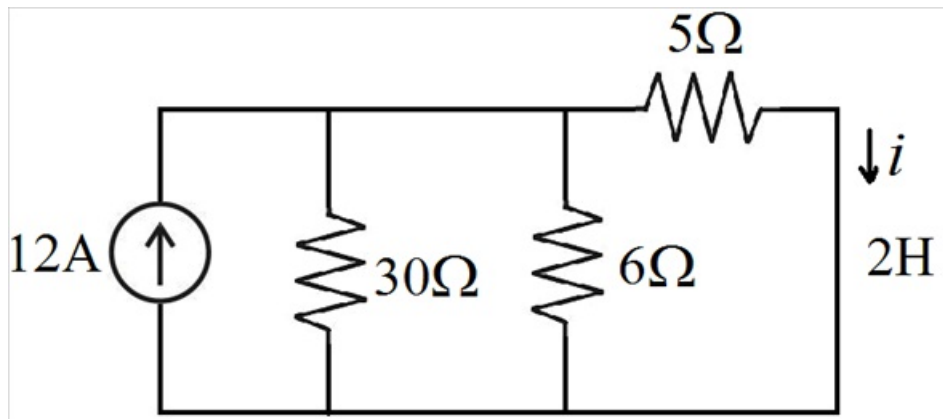
Step 3 of 13



Step 4 of 13

If the switch is closed for a long time circuit reached to steady state and inductor becomes short circuited.

Step 5 of 13



Step 6 of 13

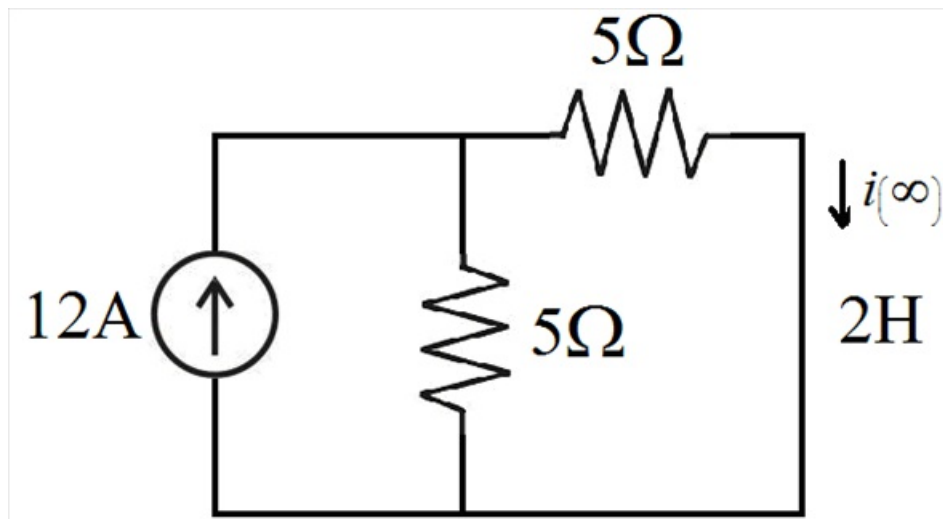
In above circuit 30Ω and 6Ω are in parallel

$$30\parallel 6\Omega = \frac{30 \times 6}{30 + 6}$$

$$30\parallel 6\Omega = 5\Omega$$

Step 7 of 13

Now the circuit becomes



Step 8 of 13

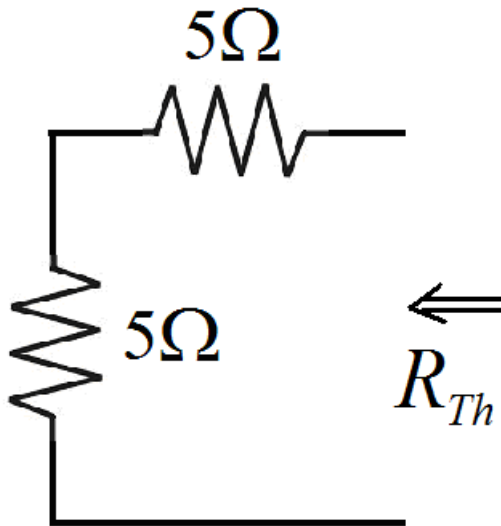
By current division rule

$$i(\infty) = 12 \times \left(\frac{5}{5+5} \right)$$

$$i(\infty) = 6\text{A}$$

For the above circuit Thevenin resistance across inductor terminals can be found by

Step 9 of 13



Step 10 of 13

$$R_{Th} = 5 + 5$$

$$R_{Th} = 10\Omega$$

$$R_{th} = 10\Omega$$

∴ Time constant

$$\tau = \frac{L}{R_{Th}}$$

$$\tau = \frac{2}{10}$$

$$\tau = \frac{1}{5} \text{ sec}$$

∴ Inductor current is

$$i(t) = i(\infty) + [i(0) - i(\infty)] e^{-\frac{t}{\tau}}$$

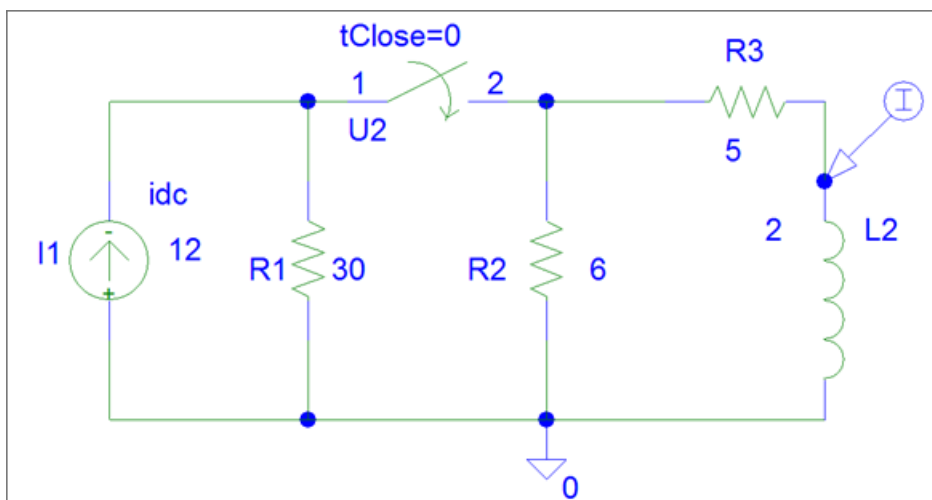
$$i(t) = 6 + [10 - 6] e^{-\frac{t}{\frac{1}{5}}}$$

$$\boxed{i(t) = 6 + 4e^{-5t} \text{ A}} \text{ for } t > 0$$

PSPICE:

Construct the PSPICE circuit as shown in figure. And double click on the inductor and give $IC = 10$ (initial current)

Step 11 of 13



Step 12 of 13

Now select analysis/setup and unmark Bias point detail and mark transient and enter final time = 2s.

After solving the circuit press F11

Then plot $i(t)$ is

Step 13 of 13

